

WORK EXPERIENCE / PROJECTS

Machine Learning / AI Internship | Computer Vision & Industrial Automation

Coogee Chemicals – Intern

Jul 2023 – Present
Melbourne, VIC

- Designed and deployed OpenCV-based real-time surveillance system using Python and YOLOv8, integrated with factory PLCs through NI-DAQmx digital/analog I/O, reducing manual monitoring labor by 70%.
- Trained deep convolutional models in PyTorch and TensorFlow to classify titanium powder samples, improving QA classification speed by 50% with 92% test accuracy.
- Optimized CNN architectures for speed and accuracy via pruning, quantization, CUDA acceleration, and low-data augmentation, reducing inference latency by 35%.
- Built end-to-end ML pipelines: dataset collection, annotation, training, and real-time deployment.
- Installed / Deployed systems across 2 remote production sites (Perth, Australia & Illinois, USA)

Capstone Project | Multi-Agent Drone Beach Patrol Simulation

University of Melbourne – Team Member

Feb 2024 – Dec 2024
Melbourne, VIC

- Developed high-fidelity drone patrol simulator in Unity HDRP with C# and Python for research into autonomous UAVs in uncertain environments.
- Integrated YOLOv8 and OpenCV for visual hazard detection with over 95% average precision across 3 object classes.
- Applied multi-agent control to coordinate drone response to dynamic threats, informed by expert surf lifesaving input.
- Emphasized sim-to-real applicability for real-time edge-case safety and generalization.
- Enhanced patrol coverage and live situational awareness for lifeguards, improving simulated response times by 40%.

Melbourne Space Program | Humanoid Robotics – Robotic Arm System

Mechatronics Engineer – Robotics Team Lead

Feb 2023 – Feb 2024
Melbourne, VIC

- Led design and closed-loop control of a 4-DOF robotic arm using ROS middleware and inverse kinematics, achieving sub-4mm positioning accuracy.
- Programmed embedded control on STM32 and Arduino using C++, integrating 12 sensors and actuators.
- Prototyped a tendon-driven soft robotic hand for adaptive gripping in humanoid applications with a 3-object grasp set.
- Managed \$10,000 budget, 6-member team, and weekly sprint design reviews with university research partners.

Mechatronics Systems Design Project | Mechanical Research Project

Team Lead

Jul 2023 – Nov 2023
Melbourne, VIC

- Directed end-to-end delivery of boring-drill prototype, from stakeholder requirements to full fabrication in 12 weeks.
- Performed component testing, safety analysis, and mechanical/electrical integration for final deployment.
- Executed full CAD-to-manufacture workflow using 3D printing, CNC, and laser in Telstra Maker Space

EDUCATION

The University of Melbourne | Ms. of Mechatronics Engineering | (3.7 GPA)

Achievements: Boeing Awards for Excellence | Capstone People’s Choice Award

Feb 2023 - May 2025
Melbourne, VIC

New York University | Exchange Ms. Mechatronics & Robotics Engineering | (3.8 GPA)

Achievements: Global Mobility Award | FEIT Exchange

Jan 2025 - Current
New York, NY

The University of Melbourne | BSc. in Mechanical Engineering | (3.8 GPA)

Achievements: Trinity Residential College Scholarship | State Volleyball Div I Champions

Feb 2020 - Dec 2022
Melbourne, VIC

The University of British Columbia | Exchange BSc. Mechanical Engineering | (3.8 GPA)

Achievements: Global Mobility Award | Intramural Volleyball Champions

Jan 2022 - Apr 2022
Vancouver, Canada

SKILLS

- Programming Languages:** Python, C++, C, Java, MATLAB, Bash Scripting
- Frameworks, Engines, OS:** ROS, NI-DAQmx, Unity HDRP, CUDA, WSL, Ubuntu Linux, OSX
- Machine Learning / AI:** OpenCV, PyTorch, TensorFlow, YOLOv8, Mask R-CNN, Ultralytics, MLOps, CI/CD
- Embedded Systems & Automation:** STM32, Arduino, PLC Integration, Real-Time Control, Digital/Analog I/O
- Design & Visualisation:** SolidWorks, AutoCAD, OnShape, CAD, 3D Printing, Laser Cutting, CNC